

Computer Science & IT

Discrete Mathematics



Combinatorics

Lecture No. 02



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Recap of Previous Lecture



Topic

Introduction to recurrence relation



Topic

Formation of recurrence relation



Topics to be Covered



★
Topic

Formation of recurrence relation

★
Topic

Solution of recurrence relation using substitution method



Topic : Formation of recurrence relation

H.W.

Q.4:-

Let a_n represents the number of n -digit binary sequences of '0' and '1' with no consecutive zeros, then

Find recurrence relation for a_n

Case ①: When first bit is '1'

n -bit



1st bit can be '1' only in one way.

$(n-1)$ bit

No restriction on 2nd bit

∴ No. of $(n-1)$ digit binary sequences with no consecutive Zero's can be given by a_{n-1}

Case ②: When first bit is '0'

n -bit



1st bit can be '0' only in one way

if 1st bit is '0', then 2nd bit must be '1'

2nd bit can be '1' only in one way

$(n-2)$ bits

No restriction on 3rd bit

∴ No. of $(n-2)$ digit binary sequences with no consecutive Zero's can be given by a_{n-2}

∴ $a_n =$ using Case ① or Case ②

$$a_n = (1 * a_{n-1}) + (1 * 1 * a_{n-2})$$

∴ $a_n = a_{n-1} + a_{n-2}$

$$a_1 = 2$$

$$a_2 = 3$$



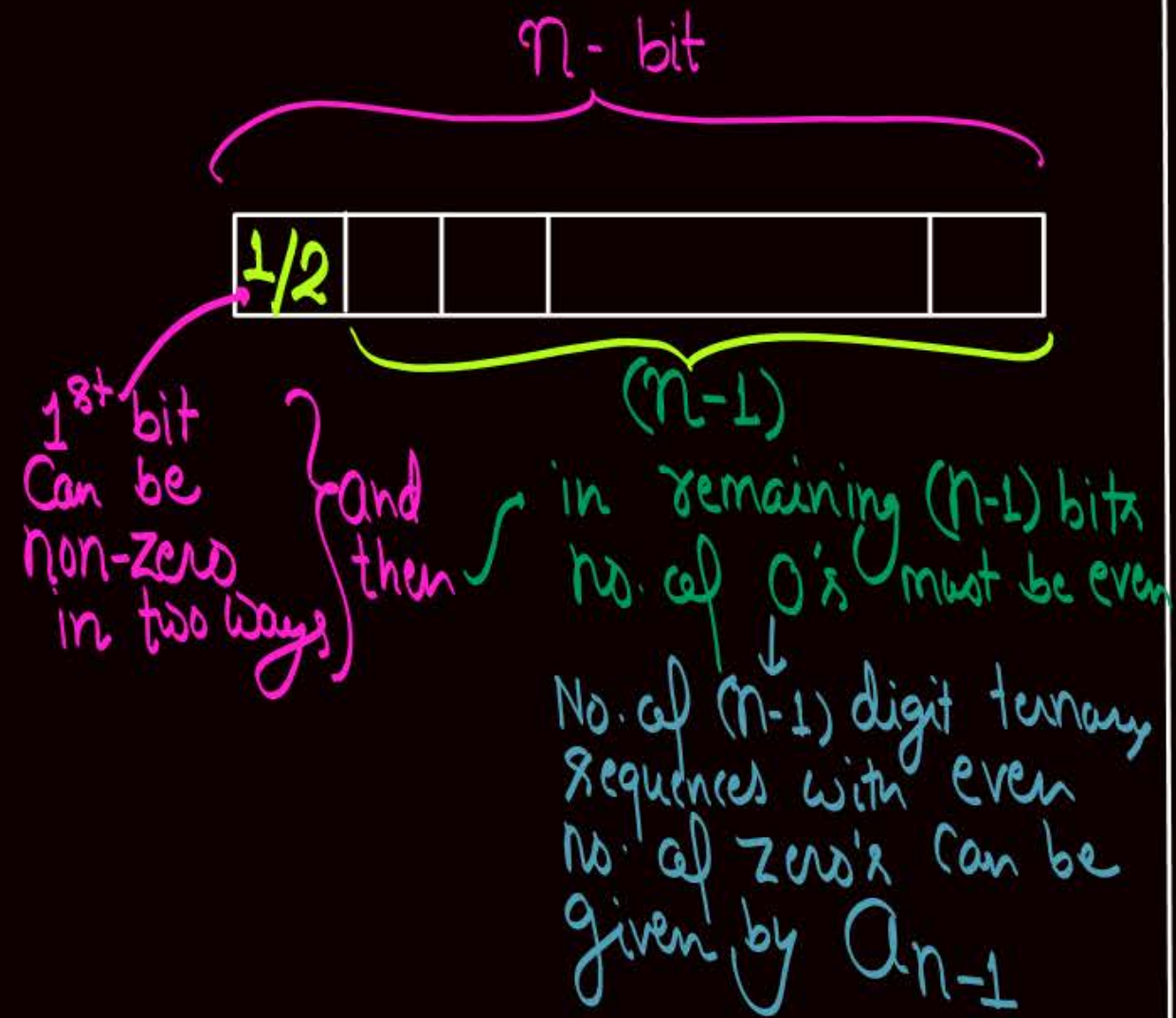
Topic : Formation of recurrence relation

H.W.

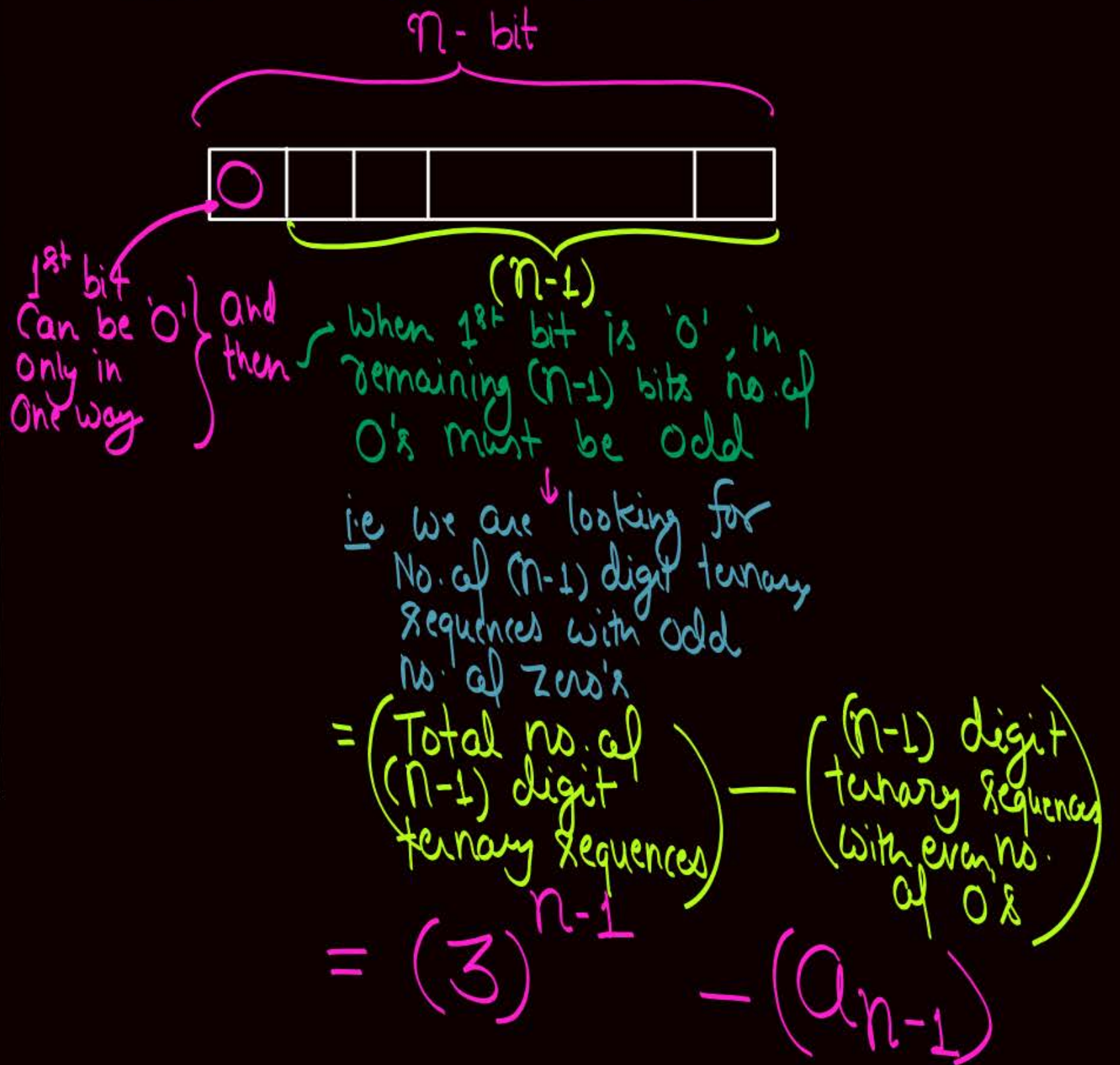
Q.5: Let a_n represents the number of n -digit ternary sequences of 0, 1, and 2 with even number of zeros in it, then

Find recurrence relation for a_n

Case ①: When first bit is not '0'



Case ②: When first bit is '0'



∴ $a_n =$ Using Case ① or Using Case ②

$$a_n = (2 * a_{n-1}) + (1 * (3^{n-1} - a_{n-1}))$$

$$a_n = 2 \cdot a_{n-1} + 3^{n-1} - a_{n-1}$$

$$a_n = a_{n-1} + 3^{n-1}$$

$$\begin{aligned} a_2 &= a_1 + 3^{2-1} \\ &= 2 + (3)^1 = 5 \end{aligned}$$

$$\begin{aligned} a_1 &= 2 \\ a_2 &= 5 \end{aligned}$$

○	○
1	2
2	1
1	1
2	2



Topic : Solution of recurrence relation

A recurrence relation is a function of the form

$$a_n = f(a_{n-1}, a_{n-2}, \dots, n)$$

→ representation of a_n as a pure function of n i.e., a function that does not contain any term of type a_{n-i} is called solution of given recurrence relation.



Topic : Solution of recurrence relation

- ✦ 1) Substitution method
- 2) Method of characteristic roots
- 3) Method of undetermined coefficients
- 4) Using generating functions



Topic : Substitution method

- In this method we use recurrence relation repetitively for $n=0,1,2,\dots$, then we solve the expression to obtain the solution of recurrence relation.

#Q. Find the solution of the recurrence relation

$$a_n = n a_{n-1}, \quad n \geq 1 \quad \text{where } a_0 = 1$$

$$a_0 = 1$$

$$a_1 = 1 \cdot a_0 = 1 \cdot 1$$

$$a_2 = 2 \cdot a_1 = 2 \cdot 1 \cdot 1$$

$$a_3 = 3 \cdot a_2 = 3 \cdot 2 \cdot 1 \cdot 1$$

$$a_4 = 4 \cdot a_3 = 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1$$

⋮

$$a_n = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = n!$$

#Q. Find the solution of the recurrence relation

$$a_n = a_{n-1} + 3^{n-1}, \quad \text{where } a_1 = 2$$

$$a_1 = 2$$

$$a_2 = a_1 + 3^1 = 2 + 3^1$$

$$a_3 = a_2 + 3^2 = 2 + 3^1 + 3^2$$

$$a_4 = a_3 + 3^3 = 2 + 3^1 + 3^2 + 3^3$$

⋮

$$a_n = 2 + 3^1 + 3^2 + 3^3 + \dots + 3^{n-1}$$

$$a_n = 2 + 3^1 + 3^2 + 3^3 + \dots + 3^{n-1}$$

$$a_n = 1 + 1 + 3^1 + 3^2 + 3^3 + \dots + 3^{n-1}$$

$$a_n = 1 + \left[1 + 3^1 + 3^2 + \dots + 3^{n-1} \right]$$

$$a_n = 1 + \left[\frac{1 \cdot (3^n - 1)}{3 - 1} \right]$$

$$a_n = 1 + \frac{3^n - 1}{2} \Rightarrow a_n = \left(\frac{3^n + 1}{2} \right)$$

it is a G.P. with
first term = 1
Common ratio = 3
No. of terms = n

$$G.P. = a + a.r + a.r^2 + a.r^3 + \dots + a.r^{n-1}$$

$\overset{\text{'''}}{a.r^0}$
 0 to $n-1$
 'n' terms

$$\text{Summation of G.P.} = \frac{a.(r^n - 1)}{(r - 1)}$$

$$\begin{aligned}
 &\text{Summation of} \\
 &\text{infinite G.P.} \\
 &= \frac{a}{1-r} \quad \text{when } |r| < 1
 \end{aligned}$$

#Q. Find the solution of the recurrence relation

$$a_n = a_{n-1} + (2n + 1), \text{ where } a_0 = 1$$

$$a_0 = 1$$

$$a_1 = a_0 + (2 \cdot 1 + 1) = 1 + (2 \cdot 1 + 1)$$

$$a_2 = a_1 + (2 \cdot 2 + 1) = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1)$$

$$a_3 = a_2 + (2 \cdot 3 + 1) = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1)$$

⋮

$$a_n = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \dots + (2n + 1)$$

$$= 1 + 3 + 5 + 7 + \dots + (2n + 1)$$

{ 0 to n, total (n+1) terms }

It is the summation of 1st (n+1) odd natural numbers

$$a_n = (n+1)^2$$

#Q. Find the solution of the recurrence relation

$$a_n = a_{n-1} + (2n + 1), \text{ where } a_0 = 1$$

$$a_0 = 1$$

$$a_1 = a_0 + (2 \cdot 1 + 1) = 1 + (2 \cdot 1 + 1)$$

$$a_2 = a_1 + (2 \cdot 2 + 1) = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1)$$

$$a_3 = a_2 + (2 \cdot 3 + 1) = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1)$$

⋮

$$a_n = 1 + (2 \cdot 1 + 1) + (2 \cdot 2 + 1) + (2 \cdot 3 + 1) + \dots + (2n + 1)$$

$$= 1 + 3 + 5 + 7 + \dots + (2n + 1)$$

$$= \frac{(n+1)}{2} [2 \cdot 1 + ((n+1) - 1) \cdot 2] = \frac{(n+1)}{2} \cdot 2 \cdot (n+1) = (n+1)^2$$

$\left. \begin{array}{l} 0 \text{ to } n, \text{ total } (n+1) \text{ terms} \\ \text{first term} = 1, \text{ Common difference} = 2 \end{array} \right\}$

Note:- ① Summation of first 'n' odd natural no. = n^2

Note: ② Consider the following A.P. series

$$a + (a+d) + (a+2d) + \dots + (a+(n-1)d)$$

First term = a ,

Common difference = d

Total no. of terms = n

$$\begin{aligned} \text{Summation of A.P. Series} &= \frac{n}{2} [2a + (n-1)d] = \frac{n}{2} \left[\underbrace{a}_{\text{first term}} + \underbrace{a + (n-1)d}_{\text{last term}} \right] \\ &= \frac{n}{2} \cdot [\text{first term} + \text{last term}] \end{aligned}$$

#Q.

Find the solution of the recurrence relation

Using Partial fraction

$$a_n = a_{n-1} + \frac{1}{n(n+1)}, \quad \text{where } a_0 = 1$$

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{(n+1)}$$

$$a_0 = 1$$

$$a_1 = a_0 + \left(\frac{1}{1} - \frac{1}{2} \right) = 1 + \left(\frac{1}{1} - \frac{1}{2} \right)$$

$$a_2 = a_1 + \left(\frac{1}{2} - \frac{1}{3} \right) = 1 + \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$a_3 = a_2 + \left(\frac{1}{3} - \frac{1}{4} \right) = 1 + \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right)$$

$$a_n = 1 + \left(\frac{1}{1} - \cancel{\frac{1}{2}} \right) + \left(\cancel{\frac{1}{2}} - \cancel{\frac{1}{3}} \right) + \left(\cancel{\frac{1}{3}} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n-1} - \cancel{\frac{1}{n}} \right) + \left(\cancel{\frac{1}{n}} - \frac{1}{(n+1)} \right)$$

$$a_n = 1 + 1 - \frac{1}{n+1} = 2 - \frac{1}{(n+1)} = \frac{(2n+1)}{(n+1)}$$



2 mins Summary



Topic

Formation of recurrence relation ✓

Topic

Solution of recurrence relation using substitution method ✓

THANK - YOU